

CO 250 Spring 2026: Assignment 3 solutions

A3-1. (a) **Variables.** There are 10 possible rectangles, which we label $j = 1, \dots, 10$. We can define a rectangle j using its bottom-left coordinates (x_j, y_j) and its length ℓ_j and height h_j (length is on the horizontal side, height is on the vertical side). We define variables x_j, y_j, ℓ_j, h_j for each j .

For a point (a_i, b_i) and a rectangle j , we define a binary variable w_{ij} to represent

$$w_{ij} = 1 \quad \Rightarrow \quad \text{point } (a_i, b_i) \text{ is inside rectangle } j$$

(Note that it is possible that the point is in the rectangle even if $w_{ij} = 0$.)

Objective function. We want to minimize the total area of the rectangles, so the objective function is

$$\min \sum_{j=1}^{10} \ell_j h_j.$$

Constraints. The lengths and heights need to be non-negative, so $\ell_j, h_j \geq 0$ for all j . We will enforce the binary condition for w_{ij} with the constraints $w_{ij}(1 - w_{ij}) = 0$ for all i, j . Notice that each equation is 0 if and only if the variable is 0 or 1.

We observe that for a point (a_i, b_i) to be in the rectangle j , it must satisfy four inequalities: $x_j \leq a_i \leq x_j + \ell_j$ and $y_j \leq b_i \leq y_j + h_j$. We can multiply w_{ij} in all the terms to get

$$w_{ij}x_j \leq w_{ij}a_i \leq w_{ij}(x_j + \ell_j), \quad w_{ij}y_j \leq w_{ij}b_i \leq w_{ij}(y_j + h_j).$$

Notice that when $w_{ij} = 1$, the inequalities all hold, so the point is in the rectangle. When $w_{ij} = 0$, we get $0 \leq 0 \leq 0$, which always holds.

We want each point (a_i, b_i) to be in at least one rectangle, so we need $\sum_{j=1}^{10} w_{ij} \geq 1$ for each i .

Full NLP.

$$\begin{aligned} \min \quad & \sum_{j=1}^{10} \ell_j h_j \\ \text{s.t.} \quad & \ell_j, h_j \geq 0 \quad \forall j = 1, \dots, 10 \\ & w_{ij}(1 - w_{ij}) = 0 \quad \forall i = 1, \dots, N, \quad j = 1, \dots, 10 \\ & w_{ij}x_j \leq w_{ij}a_i \quad \forall i = 1, \dots, N, \quad j = 1, \dots, 10 \\ & w_{ij}a_i \leq w_{ij}(x_j + \ell_j) \quad \forall i = 1, \dots, N, \quad j = 1, \dots, 10 \\ & w_{ij}y_j \leq w_{ij}b_i \quad \forall i = 1, \dots, N, \quad j = 1, \dots, 10 \\ & w_{ij}b_i \leq w_{ij}(y_j + h_j) \quad \forall i = 1, \dots, N, \quad j = 1, \dots, 10 \\ & \sum_{j=1}^{10} w_{ij} \geq 1 \quad \forall i = 1, \dots, N \end{aligned}$$

(b) For each j , we define binary variables u_j to represent

$$u_j = \begin{cases} 1 & \text{rectangle } j \text{ is used} \\ 0 & \text{otherwise} \end{cases}$$

We want that when a rectangle is not used, it is being put somewhere that never contains any point, say $(-1, -1)$ with zero length and height. So we want to have these conditions:

- When $u_j = 0$, $-1 \leq x_j, y_j \leq -1$, $\ell_j, h_j \leq 0$.
- When $u_j = 1$, $x_j, y_j, \ell_j, h_j \leq 1000$.

These conditions can be formulated using the constraints:

$$\begin{aligned} -1 + u_j &\leq x_j \leq -1 + 1001u_j \\ -1 + u_j &\leq y_j \leq -1 + 10001u_j \\ \ell_j &\leq 1000u_j \\ h_j &\leq 1000u_j \end{aligned}$$

We need the binary constraints $u_j(1 - u_j) = 0$.

We define a new variable z to represent the number of rectangles that are used. Then $z = \sum_{j=1}^{10} u_j$. We want z to be equal to 2, 3 or 7. This can be formulated using the constraint $(z - 2)(z - 3)(z - 7) = 0$ for each i . Notice that this equation holds if and only if at least one factor is equal to 0, meaning $z = 2, z = 3$ or $z = 7$, which is what we wanted.

A3-2. (a) We claim that this is unbounded. Consider the series of vectors

$$x(t) = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 5 \\ 9 \end{pmatrix} + t \begin{pmatrix} 0 \\ 3 \\ 4 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

We first check that $x(t)$ is feasible for any $t > 0$.

$$\begin{aligned}
Ax(t) &= \begin{pmatrix} 2 & 1 & 0 & 5 & -3 & 2 \\ 7 & -3 & 2 & 2 & 1 & -1 \\ 3 & 6 & -5 & -6 & 2 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 5 \\ 9 \end{pmatrix} + t \begin{pmatrix} 2 & 1 & 0 & 5 & -3 & 2 \\ 7 & -3 & 2 & 2 & 1 & -1 \\ 3 & 6 & -5 & -6 & 2 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 3 \\ 4 \\ 0 \\ 1 \\ 0 \end{pmatrix} \\
&= \begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\
&= \begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix} = b.
\end{aligned}$$

And $x(t) \geq 0$ since both vectors are non-negative vectors, and $t > 0$. So $x(t)$ is feasible. The objective value of $x(t)$ is

$$\begin{aligned}
c^T x(t) &= (-5 \ 6 \ -1 \ 3 \ -4 \ 2) \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 5 \\ 9 \end{pmatrix} + t (-5 \ 6 \ -1 \ 3 \ -4 \ 2) \begin{pmatrix} 0 \\ 3 \\ 4 \\ 0 \\ 1 \\ 0 \end{pmatrix} \\
&= -7 + 10t \rightarrow \infty
\end{aligned}$$

as $t \rightarrow \infty$. So this LP is unbounded.

For a solution with objective value is 250, we need $-7 + 10t = 250$, so $t = \frac{257}{10}$. The corresponding solution is

$$x\left(\frac{257}{10}\right) = \left(1 \ \frac{771}{10} \ \frac{514}{5} \ 0 \ \frac{307}{10} \ 9\right)^T$$

- (b) We claim that this is infeasible. We first multiply the vector $y^T = (2, -9, -5)$ on both sides of the constraints:

$$(2 \ -9 \ -5) \begin{pmatrix} 3 & 2 & 0 & 2 & 5 \\ 4 & 2 & -3 & -1 & 0 \\ -6 & -3 & 4 & 2 & 2 \end{pmatrix} x = (2 \ -9 \ -5) \begin{pmatrix} 9 \\ 2 \\ 3 \end{pmatrix}.$$

Simplifying this gives

$$(0 \ 1 \ 7 \ 3 \ 0) x = -15.$$

We see on the left hand side that we have the dot product of a non-negative vector with $x \geq 0$, so the left hand side is non-negative. However, the right hand side is negative, which is not possible. Therefore, the system is infeasible.

- (c) We claim that this LP has an optimal solution. We consider $\bar{x} = (5 \ 0 \ 0 \ 0 \ 0 \ 3)^T$. We first show that this is feasible.

It is easy to see that $\bar{x} \geq \mathbb{0}$, and

$$A\bar{x} = \begin{pmatrix} 2 & 7 & -6 & 8 & -1 & 1 \\ 3 & -5 & -6 & -1 & -4 & -4 \\ 6 & -2 & 0 & -5 & 3 & -7 \end{pmatrix} \begin{pmatrix} 5 \\ 0 \\ 0 \\ 0 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 13 \\ 3 \\ 9 \end{pmatrix} = b.$$

The objective value is

$$c^T \bar{x} = (3 \ 1 \ 4 \ -5 \ 2 \ -3) \begin{pmatrix} 5 \\ 0 \\ 0 \\ 0 \\ 0 \\ 3 \end{pmatrix} = 6.$$

We want to show that any feasible solution has objective value at most 6. We consider $y = (0, -1, 1)^T$, and multiply y^T on both sides of $Ax = b$ to get

$$(0 \ -1 \ 1) \begin{pmatrix} 2 & 7 & -6 & 8 & -1 & 1 \\ 3 & -5 & -6 & -1 & -4 & -4 \\ 6 & -2 & 0 & -5 & 3 & -7 \end{pmatrix} x = (0 \ -1 \ 1) \begin{pmatrix} 13 \\ 3 \\ 9 \end{pmatrix},$$

which simplifies to

$$(3 \ 3 \ 6 \ -4 \ 7 \ -3) x = 6.$$

Rearranging gives

$$0 = 6 - (3 \ 3 \ 6 \ -4 \ 7 \ -3) x.$$

We can add this expression to the objective function without changing its objective value. So an equivalent objective function is

$$(3 \ 1 \ 4 \ -5 \ 2 \ -3) x + 6 - (3 \ 3 \ 6 \ -4 \ 7 \ -3) x = 6 + (0 \ -2 \ -2 \ -1 \ -5 \ 0) x.$$

For any feasible solution x , $x \geq \mathbb{0}$. And since $(0 \ -2 \ -2 \ -1 \ -5 \ 0) \leq \mathbb{0}$, the value of $(0 \ -2 \ -2 \ -1 \ -5 \ 0) x$ is at most 0. Therefore, the objective value of any feasible solution is at most 6. But $\bar{x}$ has objective value exactly 6, therefore $\bar{x}$ is an optimal solution.

- A3-3. (a) We claim that if there exist a feasible solution $\bar{x}$ and a vector d such that $d \leq 0, Ad \geq 0, c^T d < 0$, then (P) is unbounded.

Let $x(t) = \bar{x} + td$ where $t > 0$. We first check that this is feasible:

$$\begin{aligned} Ax(t) &= A(\bar{x} + td) \\ &= A\bar{x} + tAd \\ &\geq b + t \cdot 0 && \text{since } A\bar{x} \geq b, Ad \geq 0, t > 0 \\ &= b \end{aligned}$$

and $x(t) = \bar{x} + td \leq 0$ since $\bar{x} \leq 0, d \leq 0$ and $t > 0$.

The objective value is

$$c^T x(t) = c^T \bar{x} + c^T(td) = c^T \bar{x} + t(c^T d).$$

We see that $c^T \bar{x}$ is a constant, and $c^T d < 0$. So as $t \rightarrow \infty$, $c^T x(t) \rightarrow -\infty$. So the LP is unbounded.

- (b) We claim that if there exists a vector y such that $y \geq 0, y^T A \geq 0^T, y^T b > 0$, then (P) is infeasible.

Suppose (P) is feasible. Then for a feasible solution x , $Ax \geq b$. We multiply both sides by y^T to get $y^T Ax \geq y^T b$, where the inequality sign remains the same since $y \geq 0$. We see that $y^T A \geq 0^T$ and $x \leq 0$, so $y^T Ax \leq 0$. However, $y^T b > 0$, so it is not possible that $y^T Ax \geq y^T b$. Therefore, (P) is infeasible.

(Note: We could have also chosen $y \leq 0, y^T A \leq 0^T, y^T b < 0$.)

- (c) If LP $\max\{c^T x : Ax \leq b\}$ has a feasible solution $\bar{x}$, then $\bar{x}$ together with $\bar{z} = 1$ forms a feasible solution for $\max\{z : Ax - bz \leq 0\}$. Moreover, for every $t \geq 0$, the solution $x = t\bar{x}$ together with $z = t$ is feasible for $\max\{z : Ax - bz \leq 0\}$, because $A\bar{x} - b \leq 0$, $t \geq 0$ and so $t(A\bar{x} - b) \leq 0$. Moreover, the solution $x = t\bar{x}$ together with $z = t$ has value t , which approaches $+\infty$ when t approaches $+\infty$.

If $\max\{z : Ax - bz \leq 0\}$ is unbounded, then it has a feasible solution $\bar{x}$ and $\bar{z}$, where $\bar{z} > 0$. From $A\bar{x} - b\bar{z} \leq 0$ and $\bar{z} > 0$, we have $A(\frac{1}{\bar{z}}\bar{x}) \leq b$. So $\frac{1}{\bar{z}}\bar{x}$ is feasible for the LP $\max\{c^T x : Ax \leq b\}$.